

Chinmoy Mitra

+88 01708770617 rudrochinmoy@gmail.com Dhaka, Bangladesh
[Portfolio](#) [LinkedIn](#) [GitHub](#) [LeetCode](#)

Education

Rajshahi University of Engineering and Technology

Bachelor of Science in Computer Science and Engineering

May 2024

Rajshahi, Bangladesh

- **CGPA:** 3.11 / 4.00
- **Relevant Coursework:** Data Structures and Algorithms (C++), OOP (C++, Java), Artificial Intelligence (Python), Neural Networks and Fuzzy Systems (Python), Digital Image Processing (Python), Linear Algebra with Computational Applications (Python)[cite: 12].

Work Experience

Dexian Bangladesh

Application Developer: AI/ML

October 15, 2025 – Present

Onsite

- Develop AI/ML-powered solutions across multiple products, including Note2Action—an automation system predicting next actions for 600+ Account Managers—and DemoFactory, an AI-driven platform that generates and deploys customizable apps from user queries.

Walton Hi-Tech Industries

AI Intern

August 2025 – October 9, 2025

Onsite

- Contributed to building a comprehensive RAG-based chatbot system for customer support, order processing, HRMS management, product search, and warranty claims, providing a one-stop solution to enhance company productivity.

Outlier

AI Contributor

March 2024 – Present

Remote

- Designed high-quality prompts and responses to improve LLM performance in code generation, refactoring, and summarization, while assisting in fine-tuning workflows through curated datasets and output evaluation.

Publications

Transfer Learning Based Multiclass Brain Tumor Classification Using MRI Data (Accepted at IEEE QPAN 2025)

[\[Paper Link\]](#)

(Undergraduate Thesis)

- Designed and implemented a Transfer Learning-based deep learning model with ResNet50 architecture for classifying brain tumor types from MRI data.
- Achieved an overall classification accuracy of 99.50% and significantly reduced computational training time.

Technical Skills

Languages

C, C++, Python, TypeScript

Machine Learning / AI

Deep Learning, NLP, RAG Systems, Transfer Learning, Transformers, LLMs, Computer Vision, Model Fine-tuning

Frameworks / Libraries

PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, DSPy, FastAPI, Hugging Face

Cloud Platforms

Microsoft Azure (Azure ML, App Service, Functions, Storage, Container Registry), Azure Foundry

DevOps / MLOps

Docker, Git, CI/CD (Azure DevOps Pipelines), API Development, Model Deployment & Monitoring

Databases

SQL, Vector Databases (FAISS, Chroma, Milvus, Cosmos DB)

Software Practices

Agile Methodology, Cloud Architecture, Data Visualization, Statistics, Requirement Analysis

Projects

Note2Action – AI-Powered Task Prediction System | *Python, Machine Learning, NLP, FastAPI*

- Built an ML system that analyzes weekly notes from 600+ Account Managers to predict next actions, reducing manual workload and minimizing missed tasks.
- Developed data processing pipelines, text classification models, and automation workflows integrated into internal AM operations.

DemoFactory – AI-driven App Generator | *RAG, LangChain, Python, Prompt Engineering, Deployment*

- Engineered an AI platform that creates customizable applications (e.g., RAG systems, summarizers) directly from user queries.
- Implemented dynamic module generation, pipeline orchestration, and automated deployment to allow instant demo creation for internal use.

Agentic AI: LLM-based Code Agent ([LINK](#)) | *OpenAI, LLama2, Python, Prompt Engineering*

- Built a multi-agent LLM system for automating coding tasks (bug fixing, test generation, code explanation) using OpenAI, LLama2, and Mixtral models.

Blog Generator via LLMs ([LINK](#)) | *LangChain, Python, Prompt Engineering*

- Designed a modular pipeline with topic prompts, outline generation, fact-checking, and rewriting agents to produce high-quality blogs.

AI Painter ([LINK](#)) | *Python, OpenCV, Mediapipe*

- Developed a gesture-controlled virtual painting app with customizable tools, real-time hand tracking, and canvas saving.

Multilingual PDF Chatbot (RAG System) ([LINK](#)) | *Python, Streamlit, FastAPI, FAISS, OCR, Gemini LLM*

- Built a RAG chatbot answering English/Bangla queries from PDFs using FAISS, Gemini LLM, OCR, Streamlit UI, and REST API.

Bangla Vowel Detection ([LINK](#)) | *Deep Learning, Transfer Learning*

- Achieved 97.76% accuracy using a transfer learning-based model to classify handwritten Bangla vowels.

4-Bit Custom CPU Design ([LINK](#)) | *Logisim, Custom ISA*

- Designed a 4-bit CPU supporting basic ALU operations and jump instructions with a custom ISA and 8x15 SRAM.

Paper Retraction Analysis with NLP ([LINK](#)) | *TF-IDF, Clustering, Random Forest*

- Applied BoW and TF-IDF to analyze retracted papers, achieving 92% classification accuracy; visualized trends via clustering.

Achievements

- Leadership Awards for being class representative for 4 consecutive years.
- Secured 10th position in RUET Codesmash Intra RUET programming contest – 2019.
- Participated in RUET GanJam Intra RUET programming contest – 2020.
- Received Dutch Bangla Bank Scholarship for Undergraduates – 2019–2023.
- Received Technical Scholarship from RUET based on Admission Merit – 2019, 2022.

References

• **Rizoan Toufiq**

Associate Professor
Rajshahi University of Engineering and Technology
Email: rizoantoufiq@cse.ruet.ac.bd
Phone: +880-1768496122

• **Farjana Parvin**

Assistant Professor (Thesis Supervisor)
Rajshahi University of Engineering and Technology
Email: farjana@cse.ruet.ac.bd
Phone: +880-1739497360